

Empiricist

14+3 · mix cards · scene-N · chunk smoke

Hypotheses → mix matrix → P0/P1 GPU table → horizon sweep → GPU/disk honesty → directional SR bar (not household Fig.8 parity).

Spine: Same Franka · many scenes · protocol > vibes · Fig.10 before Fig.8 · 2-16-8 reactive · steal protocol into BEHAVIOR

Success = loader + ≥ 2 mix cards with directional SR/MSE + GPU/disk/scene logs

Hypotheses under test

From DELIVERABLE / pack

H-droid-lift

50/50 DROID-subset co-train > none on ID and especially OOD (Fig. 8 moral)

H-scene>volume

At fixed N, high scene-count subsample > top-scenes on OOD (Fig. 10)

H-vs-oxe

On Franka-like proxies, DROID diversity can beat heterogeneous OXE of similar/larger N

H-chunk-horizon

pred 16 / exec 8 @ ~15 Hz improves multi-stage completion vs single-step BC

H-fail / lang / protocol

Falsifiable variants: include fails; multi-lang; protocol ablations (DELIVERABLE §3)

Minimal mix matrix

$\{\text{mix}\} \times \{\text{init}\} \times \{\text{FT}\}$

mix $\in \{ \text{none} \mid \text{droid_diverse_N} \mid \text{droid_top_scenes_N} \mid \text{oxe_subset} \}$

init $\in \{ \text{scratch_DP_small} \mid \text{octo} \mid \text{openvla-lora} \}$ · FT = tiny in-domain · + one chunk {8,16} contrast

none

In-domain only baseline

Fig. 8 No Co-train

droid_diverse_N

Uniform scene sample @ N

Fig. 10 diverse arm

droid_top_scenes_N

Top-K scenes @ matched N

Fig. 10 20-scenes arm

oxe_subset

Octo-soup – Lang Table style

Fig. 8 OXE arm

P0 / P1 GPU table

Smoke-first · outlines in env_outline/

Pri	Experiment	Smoke	GPU-h
P0	Shard inventory + schema smoke	Y	<1
P0	Co-train mix card none / DROID / OXE	Y	8-24
P0	Scene-diversity matched-N (Fig. 10)	Y	10-30
P1	Horizon/chunk {8,16,32}×{4,8}; fail; lang; cam-aug	Y/Partial	6-24
P1	Same mix → Octo / OpenVLA-LoRA / π_0	Partial	20-60
Defer/N	Full 76k DP × multi-site; Cook Lentils parity; 0-demo ZS	N	>>200

Horizon / chunk deep dive

$\text{pred} \in \{8, 16, 32\} \times \text{exec} \in \{4, 8\} @ \sim 15 \text{ Hz}$

Paper recipe: $\text{obs}=2$ / $\text{pred}=16$ / $\text{action}=8 \rightarrow \text{commit_s} = 8/15 \approx 0.53 \text{ s}$

Reactive chunked BC — NOT classical planner. Log commit_s next to every SR.

pred=8 exec=8

$\text{commit_s} \approx 0.53 \text{ s}$

pred=16 exec=8

$\text{commit_s} \approx 0.53 \text{ s}$

PAPER DEFAULT

pred=32 exec=8

$\text{commit_s} \approx 0.53 \text{ s}$

pred=8 exec=4

$\text{commit_s} \approx 0.27 \text{ s}$

pred=16 exec=4

$\text{commit_s} \approx 0.27 \text{ s}$

pred=32 exec=4

$\text{commit_s} \approx 0.27 \text{ s}$

Long-horizon proxy — Cook Lentils / Clean Desk

Stage completion metric, not household parity required

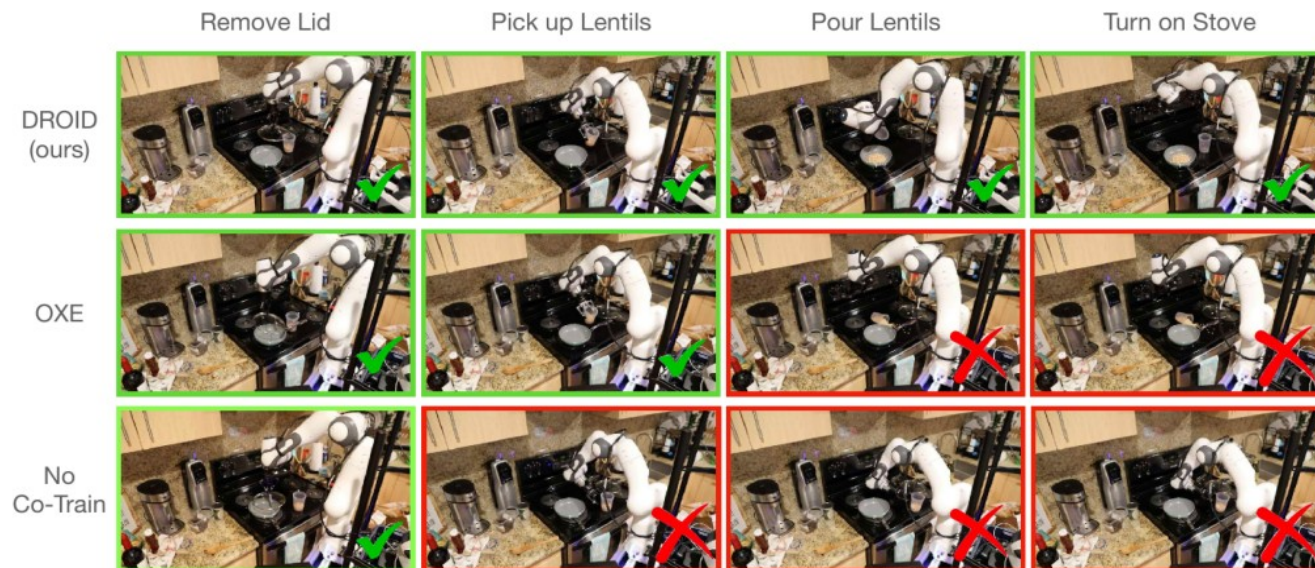


Fig. 9: Representative policy rollout examples on the most challenging “Cook Lentils” task in the kitchen of one of the authors. Qualitatively, we find that policies co-trained with DROID perform smoother, more precise motions, allowing them to solve long-horizon tasks like lentil cooking even in the presence of unseen distractor objects. In contrast, policies trained only with in-domain demonstrations or co-trained with Open-X data [39] struggle with long-horizon tasks and out-of-distribution evaluation settings. See <https://droid-dataset.github.io> for rollout videos.

Empiricist bar

- Metric: stages completed / episode
- Compare chunk vs single-step BC
- OOD: distractors + camera shift
- Directional lift is enough
- Do NOT claim Fig. 8 household parity
- without robots + full protocol
- 50 ID demos · 50/50 mix card

P0 smoke — Fig. 10 matched-N

diverse_N > top_scenes_N on OOD proxy

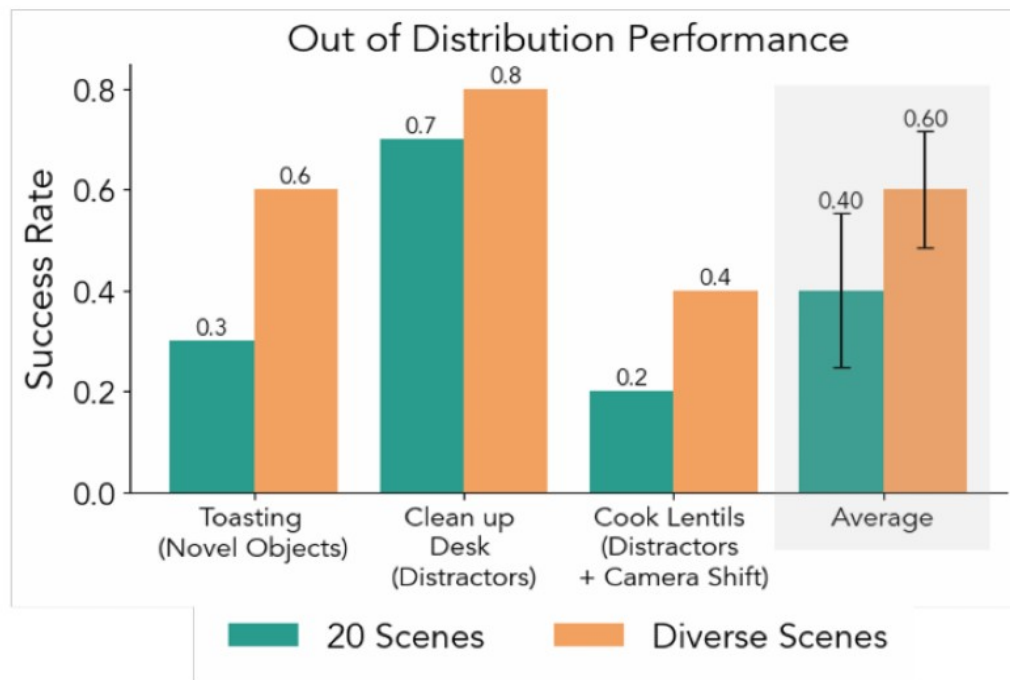


Fig. 10: **How important is the scene diversity in DROID?** We find that co-training on a subset of DROID with diverse scenes has higher OOD performance than co-training on a subset of DROID with only 20 scenes, suggesting the scene diversity of DROID is one of the

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Run card

- Fix N (e.g. 2k-7k scout)
- Build diverse vs top-K scenes
- Same init + tiny ID FT
- Log scene cardinality
- Log mix card hash
- Directional OOD SR enough
- GPU-h: 10-30 scout

P0 smoke — Fig. 8 mix card

none vs DROID vs OXE · directional only



Fig. 8: Does DROID Improve Policy Performance and Robustness? We find that across all our evaluation tasks, co-training with DROID significantly improves both in distribution and OOD performance over both no co-training and co-training with the Open-X dataset. We compare success rate averaged across all tasks with standard error, and find DROID outperforms the next best method by 22% absolute success rate in-distribution and by 17% out of distribution.

In Figure 10 we observe that using the split of the dataset with more diverse scenes yields better performance in the OOD evaluation setting. By comparing Figure 10's individual task performances with the corresponding tasks in Figure 8, we also see that the performance of co-training with the full DROID dataset matches or outperforms the performance with the subsampled dataset on all three tasks. These results suggest that the strength of DROID lies in its size and especially in

We hope that DROID can be a catalyst for research on general-purpose robot manipulation policies that are able to generalize to a broad range of tasks and scenes. In this work, we showed one example for leveraging DROID to boost policy performance, but there are many open questions about how to best make use of such diverse data: how should we combine DROID with existing large-scale robot datasets and how can we train policies that perform tasks in new scenes without any

Honesty

- Target: DROID-subset > none
- on ONE OOD proxy
- Not +22/+17 across 6 tasks
- Pin ~40k vs 76k in logs
- OXE = curated soup, say so
- 10 rollouts ok for smoke
- SE / CI if claiming publish

GPU / disk honesty

TB-class full corpus = N · subsets only

Workload	VRAM	Disk	Wall	Smoke
Shard schema smoke	0–8 GB	20–200 GB selective	min–hours I/O	Y
Small DP co-train scout	12–24 GB	+ ckpt	4–12 h	Y
Scene-diversity pair	12–24 GB	shared	1–2 days	Y
OpenVLA-7B LoRA + mix	24–48 GB	+15–30 GB ckpt	8–24 h	Partial
Chunk sweep ×3	12–24 GB	shared	1–2 days array	Y
Full DROID mirror / multi-site A/B	—	≫1–several TB + robots	—	N

Success bar vs paper claims

Directional SR + logs — NOT household parity

Usable in-the-wild Franka data

✓ Feasible: Loader streams multi-cam / action / lang / success

✗ Not required: Full 76k ingest

Co-train ↑ robustness

✓ Feasible: DROID-subset > none on one OOD proxy

✗ Not required: +22/+17 across 6 real tasks

Scene diversity drives OOD

✓ Feasible: diverse-N > top-scenes-N @ matched N

✗ Not required: Exact Fig. 10 bars

Chunk helps long-horizon

✓ Feasible: Chunked ↑ stage completion on one multi-stage proxy

✗ Not required: Household Cook Lentils robot

Env outline pointers

Outlines only — not sbatch'd from quiet box

env_outline/conda_env.yaml

mamba sketch

env_outline/setup_mamba.sh

env create + subset download pointers

env_outline/slurm_shard_smoke.sh

1-shard schema inventory

env_outline/slurm_cotrain_mix.sh

none / DROID / OXE mix cards

env_outline/slurm_scene_diversity.sh

diverse-N vs top-scenes-N

env_outline/slurm_horizon_chunk.sh

pred/exec sweep

env_outline/run_checklist.md

GPU/disk honesty preflight

Protocol ablations to log

When claiming H-protocol

Scene time-cap

≤ 20 min / ~ 100 traj vs unlimited

GUI random vs free choice

Task sampling bias control

View/lighting jitter

Cam nudge + lighting augs on/off

Fail include vs exclude

$\sim 16k$ fails in or out of mix

Lang multi ≤ 3

Single vs multi-instruction

Wrist on/off

Paper default = exterior only

Numbers tattoo — pin these

Shared across roles · Nov 6 ready

76k

successful traj

350 h

teleop hours

564

scenes

52

buildings

86

task verbs

50

collectors

13 / 18

labs / robots

+22% ID

Fig. 8 lift

+17% OOD

Fig. 8 lift

7362

Fig. 10 matched N

2 / 16 / 8

obs/pred/act @15Hz

~40k ≠ 76k

co-train pool

Empiricist — close

Takeaways + exam bite

1. Run P0: schema smoke + mix card + matched-N scene diversity before any architecture claim.
2. Horizon sweep pred×exec; always log commit_s next to SR.
3. Success = directional lift + GPU/disk/scene logs — never claim household Fig. 8 parity from a scout.
4. TB-class full corpus = N; prefer HF resized mirrors; pin 40k≠76k in every mix card.

EXAM BITE

H-scene>volume @ matched N; 2-16-8 commit≈0.53s; directional SR only.